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# Introduction to Cancer Genomics & MSK-IMPACT

GEMS Training

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# Agenda

- Genomics
  - Cancer genomics
  - Precision oncology
  - Genetic mutations
  - Genomic sequencing
- MSK–IMPACT
  - MSK genomics
  - IMPACT data – {cbioportalR}
  - IMPACT analyses and visualizations – {gnomeR}



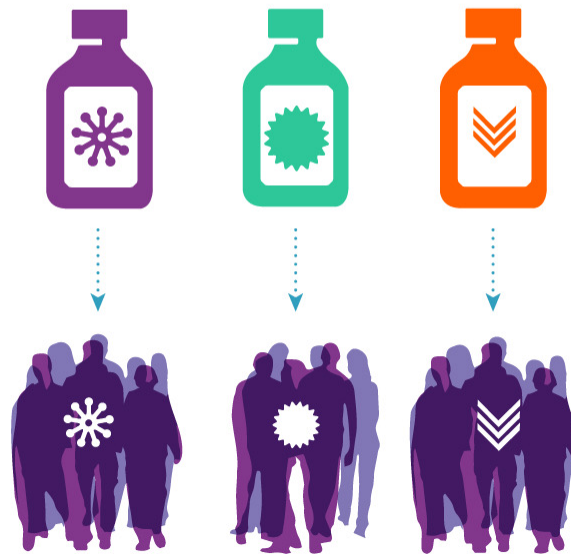
# Cancer is a Genetic Disease



- Cancer is a collection of diseases characterized by uncontrolled cell growth
- Cancer cells can divide rapidly, evade protective mechanisms that kill abnormal cells, and metastasize
- These changes are caused by a series of somatic mutations

## UNDERSTANDING PRECISION MEDICINE

In precision medicine, patients with tumors that share the same genetic change receive the drug that targets that change, no matter the type of cancer.



Credit: National Cancer Institute



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# What is Precision Oncology?



National Human Genome  
Research Institute

Est 1989

*"An emerging medical discipline that involves using genomic information about an individual as part of their clinical care (e.g., for diagnostic or therapeutic decision-making) and the health outcomes and policy implications of that clinical use."*



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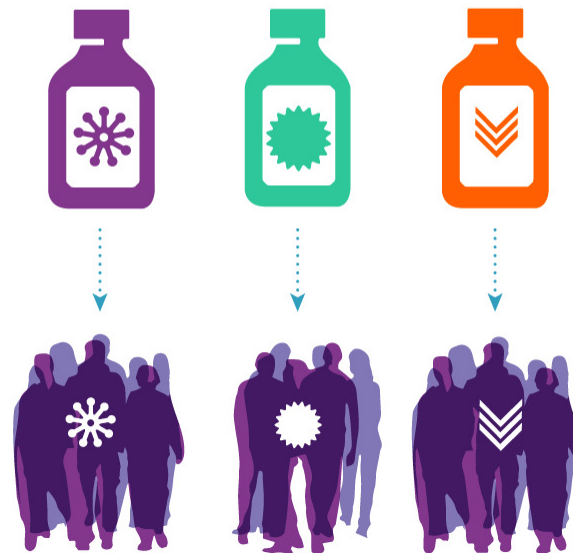
# What is Precision Oncology?



*Genomic information about cancer is leading to better diagnoses and treatment strategies that are tailored to patients' tumors, an approach called **precision medicine**.*

## UNDERSTANDING PRECISION MEDICINE

In precision medicine, patients with tumors that share the same genetic change receive the drug that targets that change, no matter the type of cancer.



Credit: National Cancer Institute



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# Cancer genomics success stories – targeted therapies

HER2+  
Breast Cancer  
“worst to first”



Trastuzumab (Herceptin) controls the hyper-active signaling pathway (HER2 tyrosine kinase) caused by multiple copies of the *HER2* gene on chr 17 in a subtype of breast cancers. (~1998)

BCR-ABL+  
Myeloid Leukemia



Chromosomal rearrangement creates oncogene BCR-ABL in Chronic Myeloid Leukemia. Imatinib (Gleevec) inhibits overactivity of the Bcr-Abl tyrosine kinase protein. (~2001)

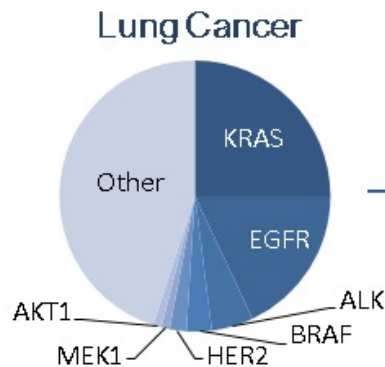
EGFR+  
Non Small Cell  
Lung Cancer



Erlotinib restricts activation of a protein (epidermal growth factor, or EGFR), which is mutated and abnormally active in a subset of lung cancers. (~2004)

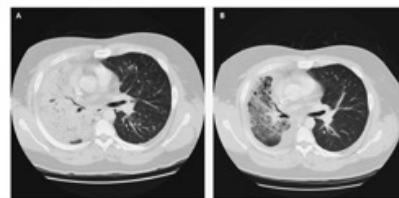
# Cancer genomics success stories – targeted therapies

## Precision Oncology

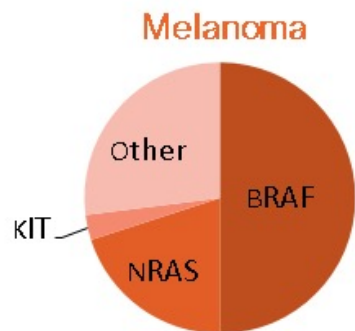


**EGFR-mutant cancer**

EGFR inhibitors:  
erlotinib, gefitinib

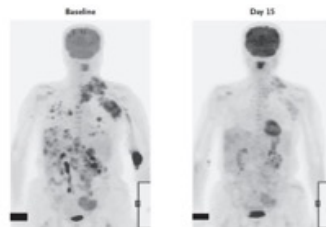


*Lynch et al., NEJM, 2004*  
*Paez et al., Science, 2004*  
*Pao et al., PNAS, 2004*



**BRAF-mutant cancer**

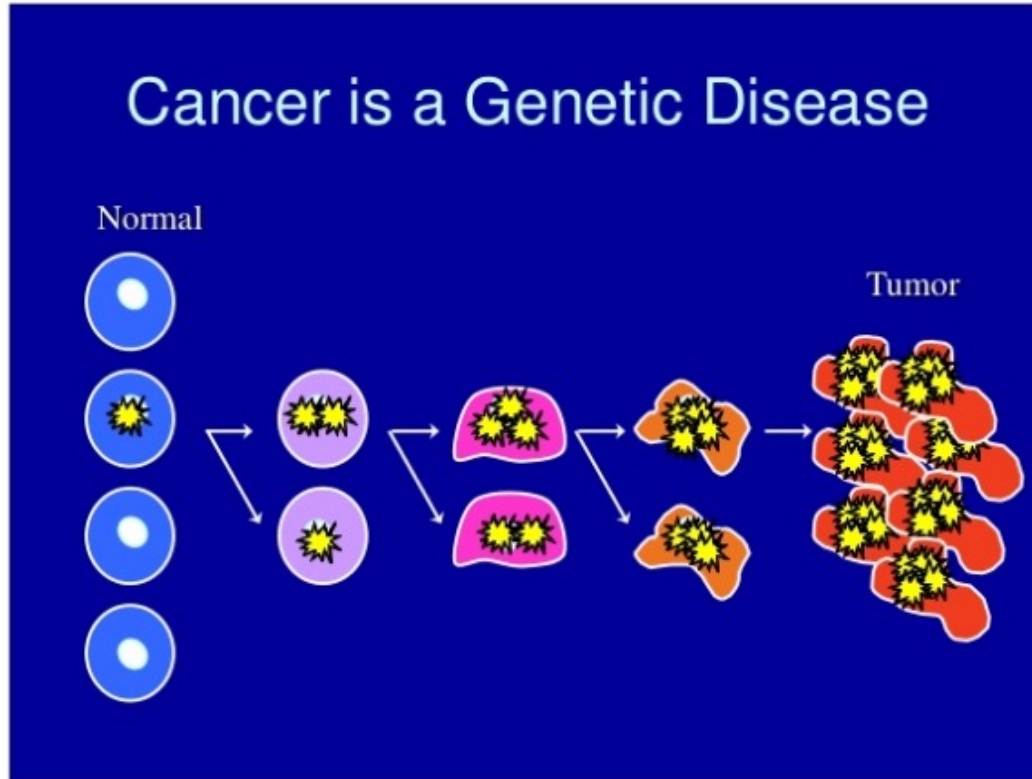
BRAF inhibitor:  
vemurafenib, dabrafenib



*Flaherty et al., NEJM, 2010*

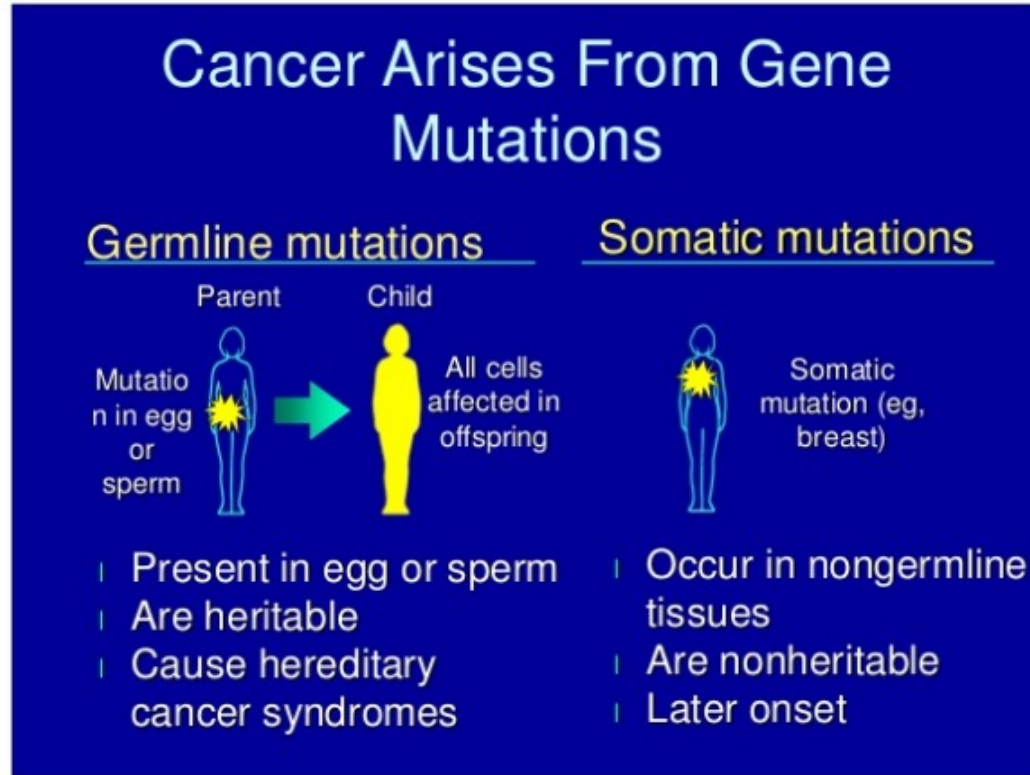


# Cancer genomics – genetics review



*The transition from a normal cell to malignant cancer is driven by DNA mutations and alterations.*

# Cancer genomics - genetics

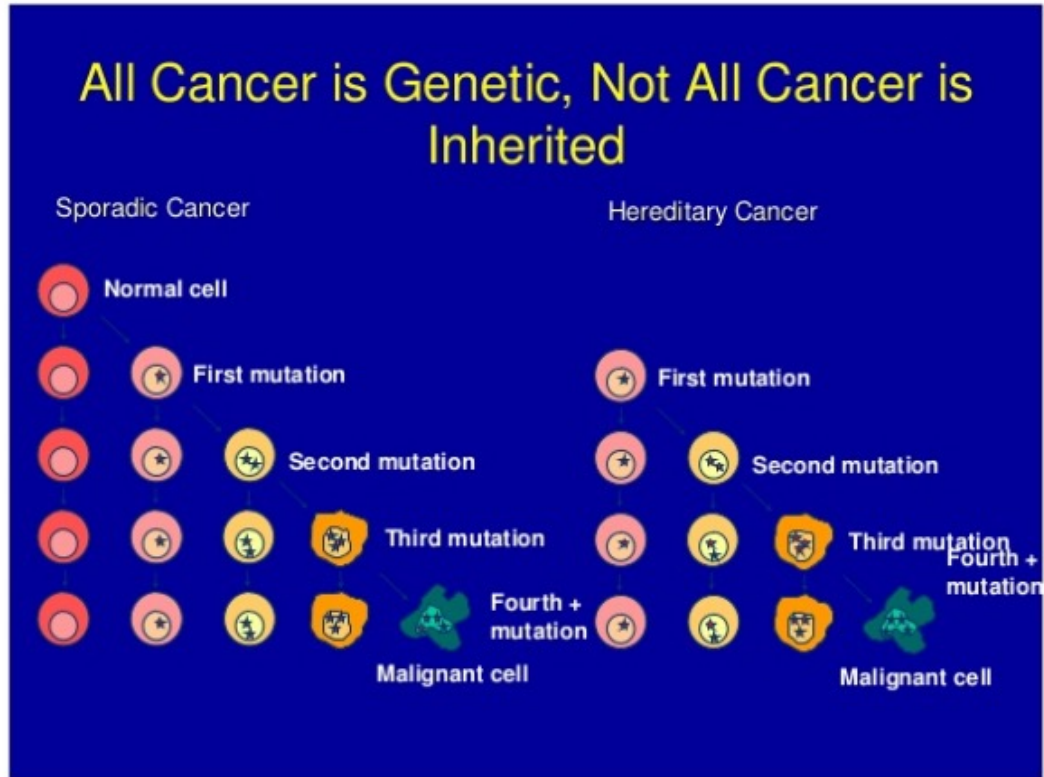


From Hereditary Cancer Predisposition, Slides by Patricia Ganz, MD, UCLA



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# Cancer genomics - genetics



From Hereditary Cancer Predisposition, Slides by Patricia Ganz, MD, UCLA

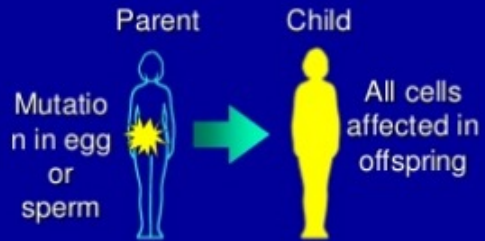


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# Cancer genomics – genetics

## Cancer Arises From Gene Mutations

### Germline mutations



- | Present in egg or sperm
- | Are heritable
- | Cause hereditary cancer syndromes

### Somatic mutations



- | Occur in nongermline tissues
- | Are nonheritable
- | Later onset

5 -10% of cancers

90 - 95% of cancers

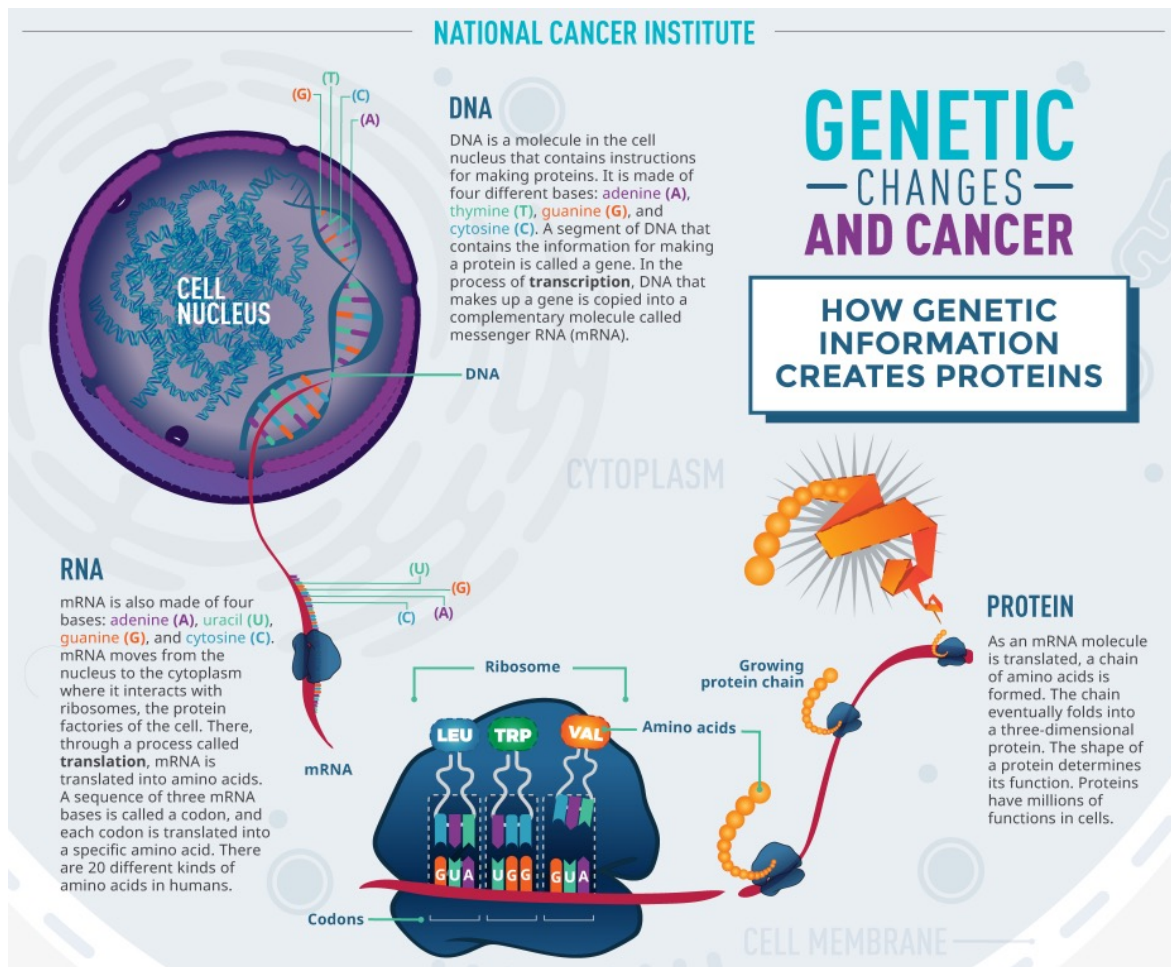
# Cancer genomics – somatic mutations

## Possible causes:

- Spontaneous occurrences at low frequency
- Errors that occur as cells divide
- Exposure to carcinogenic substances, e.g.
  - Radiation (UV rays from sun)
  - Chemicals in tobacco smoke



# Cancer genomics – somatic mutations



*How a mutation causes damage:*

*DNA to RNA to Protein*

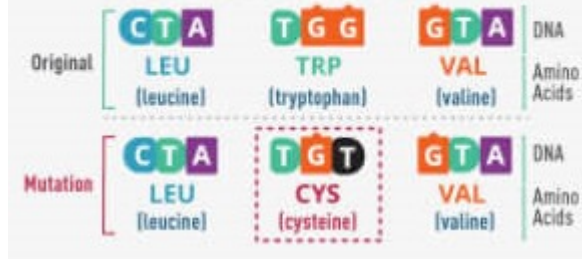


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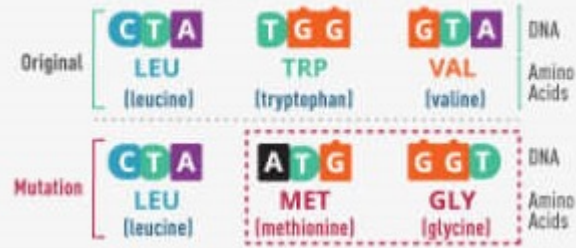
# Cancer genomics – somatic mutations

## TYPES OF GENETIC MUTATIONS IN CANCER

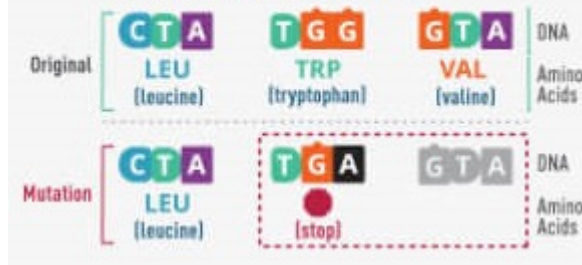
### MISSENSE MUTATION



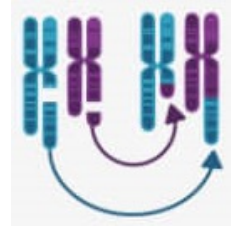
### FRAMESHIFT MUTATION



### NONSENSE MUTATION



### CHROMOSOME REARRANGEMENTS



*Mutations can change proteins, which can change cells behavior...*

# Cancer genomics – detecting somatic mutations

## Tumor DNA sequencing assays

- Tumor sample from biopsy or resection
- Compares DNA sequence in cancer cells with that in normal cells
- Broad or targeted focus
  - Broad regions: whole genome / whole exome sequencing
  - Targeted regions: gene panels, e.g. MSK-IMPACT





# MSK IMPACT

## Integrated Mutation Profiling of Actionable Cancer Targets

*Goal: Expanding genomic testing to more patients*

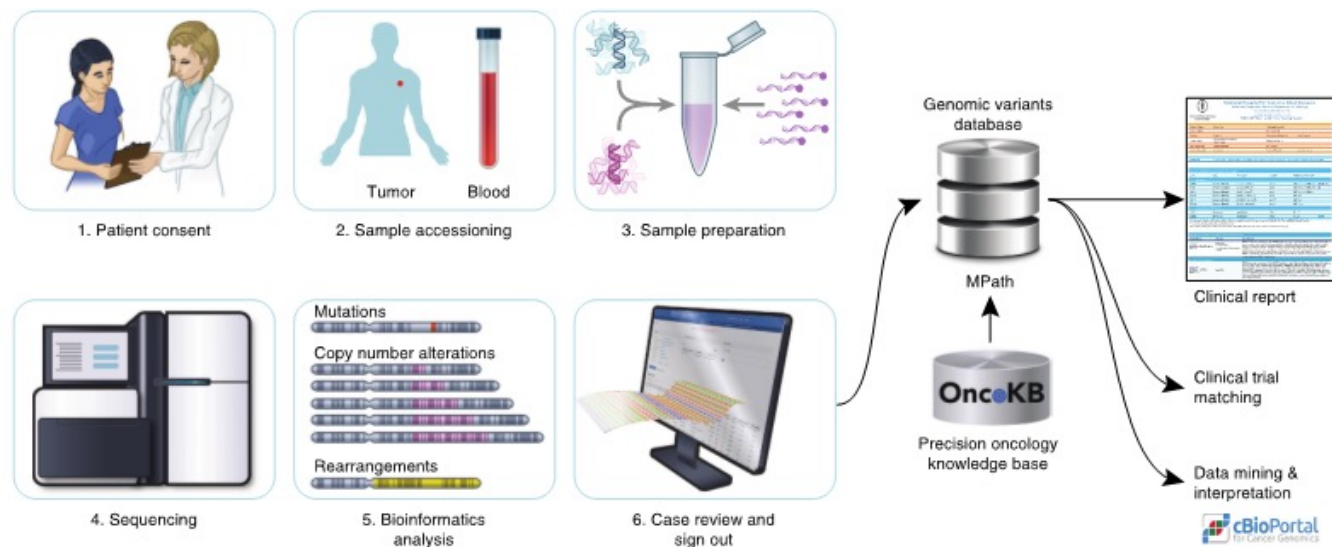
- Now routinely used in clinical practice to sequence patient tumors.
- Nov 2017: MSK-IMPACT was first tumor-profiling test to be authorized by US FDA



# MSK IMPACT

## Precision Oncology with MSK-IMPACT

Identifies **mutations**, **copy number alterations**, **structural variants** in **468 genes** as well as biomarkers that could predict treatment response such as **MSI status** or **tumor mutation burden (TMB)**



# MSK IMPACT

## MSK-IMPACT Genes

Panel includes exons, introns, and other noncoding regions

**22,118 cases**

341 genes: 2,861 cases

**410 genes: 10,004 cases**

**468 genes: 9,253 cases**

### Cancer Gene Exons:

**6,614 protein-coding exons of 468 genes**

- actionable mutations
- targets of investigational agents
- frequently mutated in cancer
- cancer susceptibility genes

### Cancer Gene Introns:

**70 introns of 20 recurrently rearranged genes**

**Cancer Gene Promoters:** TERT promoter

**SNP Probes:** >1,000 non-coding SNPs

- copy number analysis, various QC checks

**Target Territory = 1.52 Mb**

# MSK IMPACT

## MSK-IMPACT Genes

Panel includes exons, introns, and other noncoding regions

22,118 cases

341 genes: 2,861 cases

410 genes: 10,004 cases

468 genes: 9,253 cases



**Updated as of 7/22:**

**~80,652 cases:**

468 genes: 36,636 cases

505 genes: 17,324 cases

### Cancer Gene Exons:

6,614 protein-coding exons of 468 genes

- actionable mutations
- targets of investigational agents
- frequently mutated in cancer
- cancer susceptibility genes

### Cancer Gene Introns:

70 introns of 20 recurrently rearranged genes

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# MSK IMPACT - OncoKB

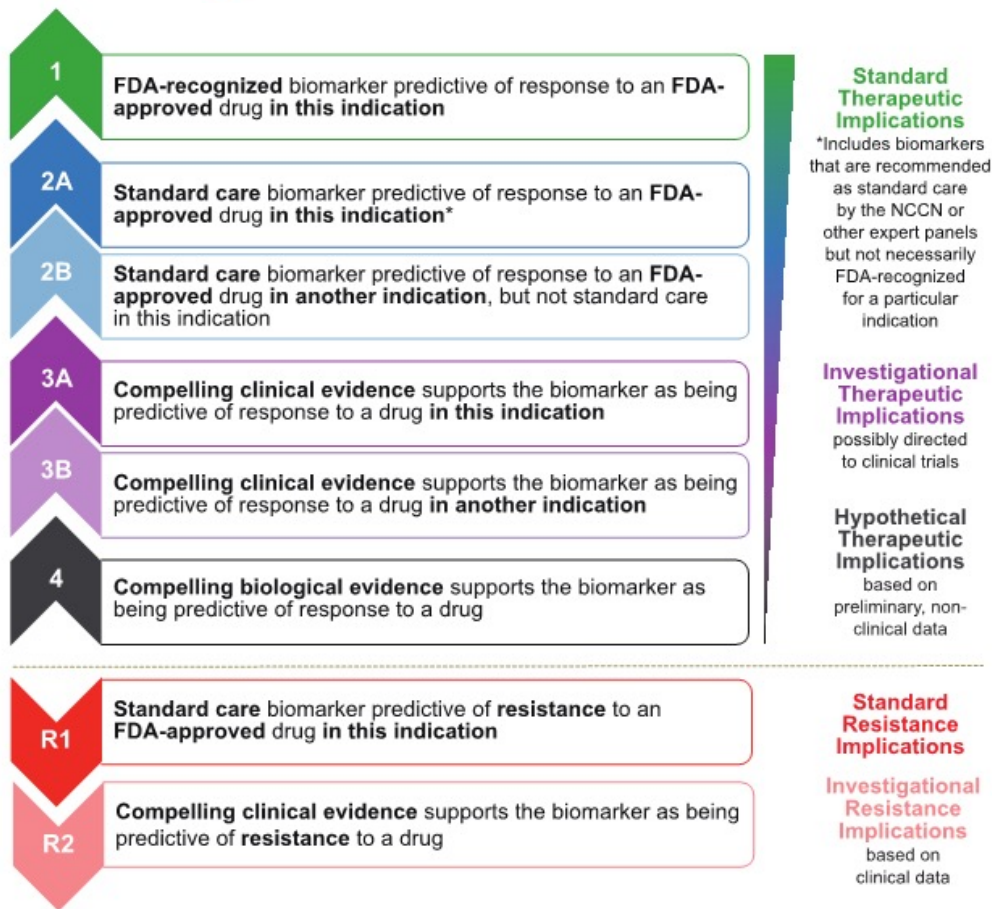
*Goal:*

*Help doctors find best therapies based on tumor's genomic profile*

OncoKB – Oncology Knowledge Base

*Contains info about likely oncogenicity and clinical actionability of specific cancer gene alterations*

## OncoKB Therapeutic Levels of Evidence

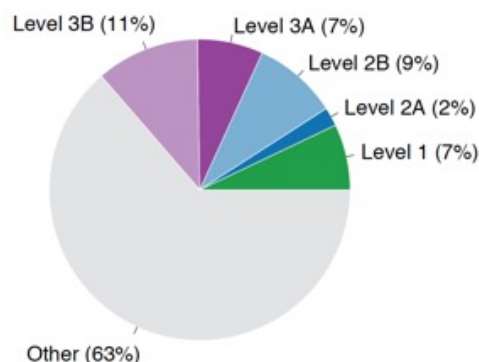


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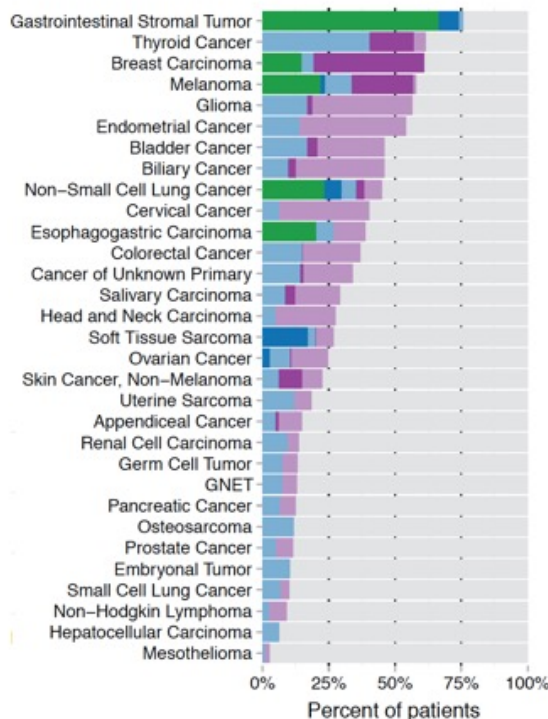
# MSK IMPACT - OncoKB

## Prevalence of Actionable Mutations

37% of patients harbored  $\geq 1$  OncoKB-annotated actionable mutation



|          |                                                                                                                   |
|----------|-------------------------------------------------------------------------------------------------------------------|
| Level 1  | FDA-recognized biomarker for an FDA-approved drug in the same indication                                          |
| Level 2A | Standard of care biomarker for an FDA-approved drug in the same indication                                        |
| Level 2B | Standard of care biomarker for an FDA-approved drug in another indication                                         |
| Level 3A | Compelling clinical evidence supporting the biomarker as being predictive of drug response in the same indication |
| Level 3B | Compelling clinical evidence supporting the biomarker as being predictive of drug response in another indication  |



# MSK IMPACT – Data sharing

*Goal:*

*Gather data and share analyses within MSK and with the public.*



- MSK-IMPACT results stripped of patient-identifying info available for scientific use by everyone at MSK through [cBioPortal](#) and by greater scientific community through AACR.
- cBioPortal provides **visualization, analysis** and **download** of published data sets.
- As of the end of 2018, more than 100 scientific publications by MSK authors have included results from MSK-IMPACT testing.



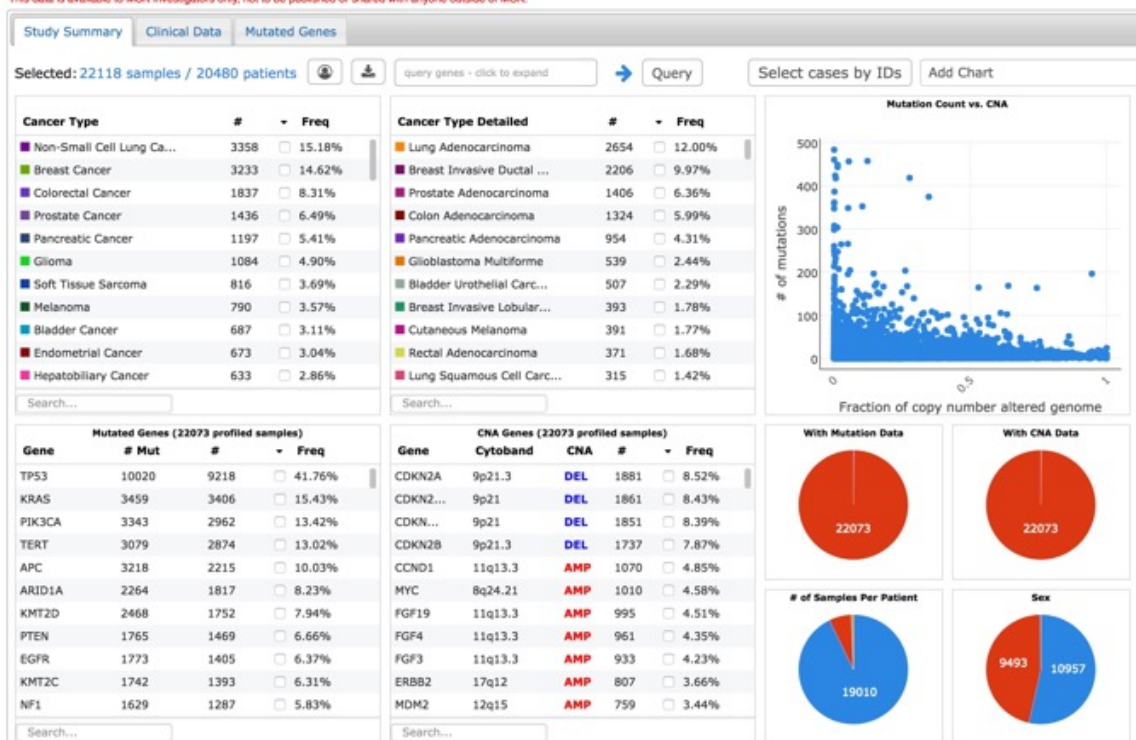
# MSK IMPACT – Data sharing

## Data sharing with Researchers

### Cohort level data

MSK-IMPACT Clinical Sequencing Cohort (MSKCC) [Query this study](#)

Targeted sequencing of clinical cases via MSK-IMPACT. Please follow the [publication guidelines](#) when using this data in abstracts or journal articles.  
This data is available to MSK investigators only, not to be published or shared with anyone outside of MSK.



# MSK IMPACT – Data for Analysis



- Data from studies in progress are also deposited in cBioPortal and investigators receive specific access to these studies
- Questions of interest:
  - Genes / mutations / pathways associated with prognosis / prediction
  - Genes / mutations / pathways associated with clinical characteristics, e.g. Stage, grade, lymphovascular invasion, perineural invasion, etc
  - Is there co-occurrence or mutual exclusivity among genes in the cohort?
  - Are there clinically meaningful subtypes in the cohort?
- Approaches to obtaining IMPACT data:
  - Ask for copy of **annotated** cBioPortal data
    - Essential if you want to look at location of mutation or **oncogenicity**, among other characteristics.
  - Download data yourself with **cbioportalR** package



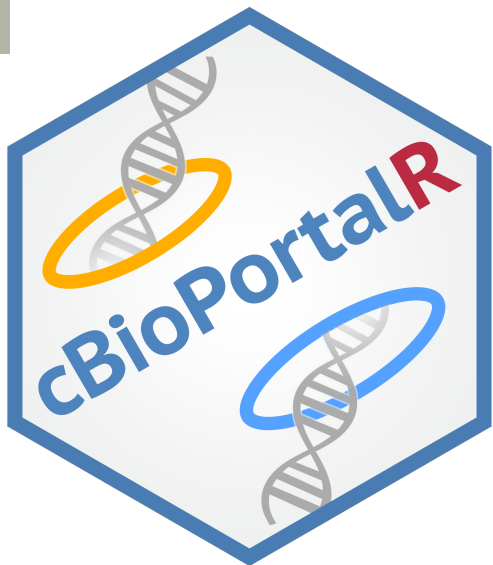
# MSK IMPACT – Data for Analysis



- Obtaining IMPACT data:
  - Download data yourself with **cbioportalR** package
  - Either with **sample ids** or **study id**



# MSK IMPACT – cbioportalR

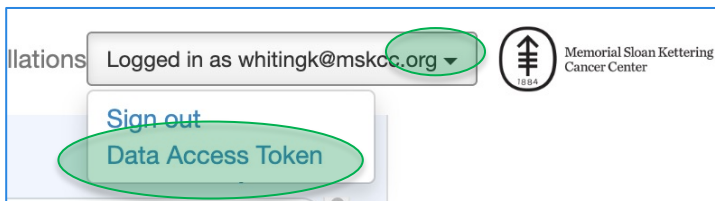


{cbioportalR} allows you to access [cBioPortal's](#) genomic and clinical data sets directly through R

- **Install the most up to date version:**  
`remotes::install_github("karissawhiting/cbioportalR")`



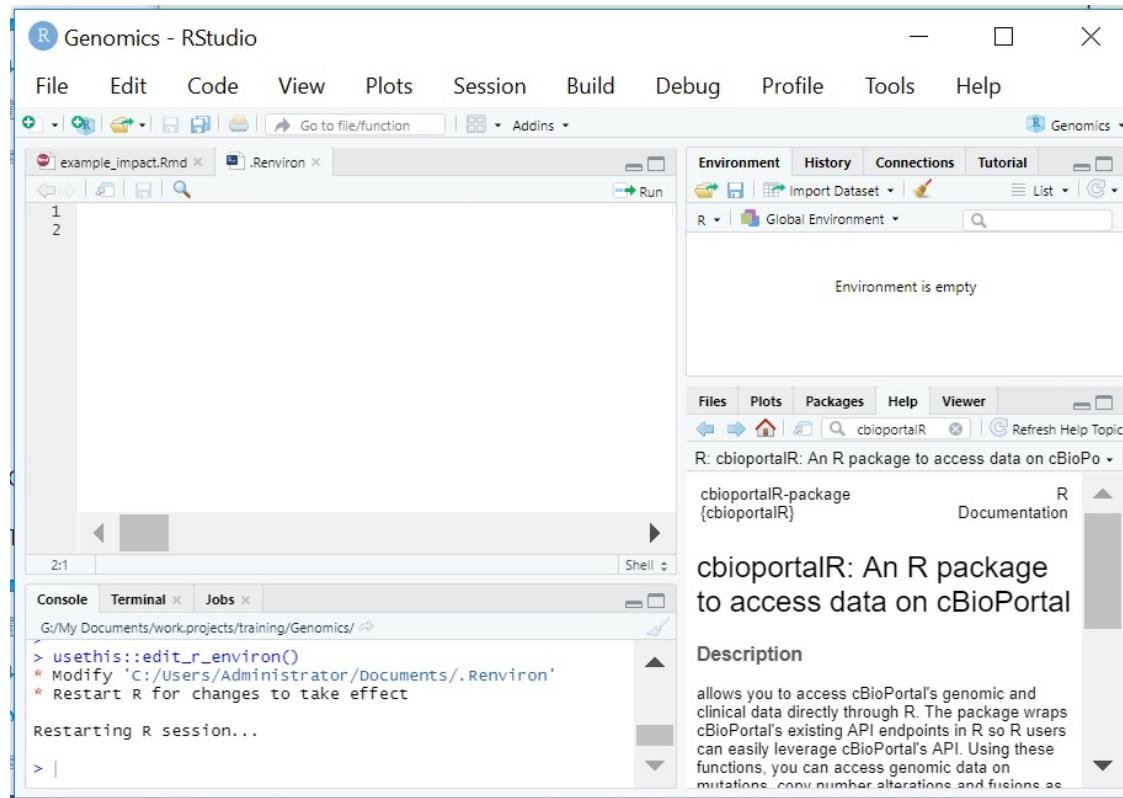
# MSK IMPACT – cbioportalR



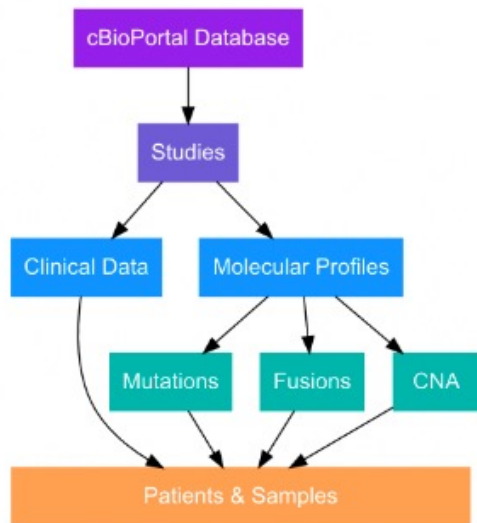
(more detailed instructions on website)

- You can access public data anytime without credentials:
  - `set_cbioportal_db("public")`
- For projects you will need to authenticate yourself to use internal MSK Data:
  - Log in and get a token from <https://cbioportal.mskcc.org>
  - Save it to your R environment  
`usethis::edit_renviro()`
-  Reload R, then: `set_cbioportal_db("msk")`

# MSK IMPACT – cbioportalR



# MSK IMPACT – How Data is Organized in cBioPortal

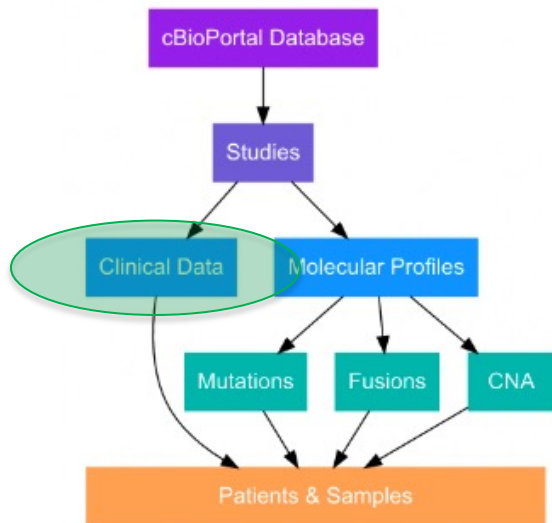


| Level              | Functions                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Database           | <code>`available_studies()`</code> , <code>`available_profiles()`</code> , <code>`available_gene_panels()`</code> , <code>`get_genes()`</code>                                                                                                                                                                                                                                                            |
| Studies            | <code>`available_profiles()`</code> , <code>`get_study_info()`</code> , <code>`available_clinical_attributes()`</code> , <code>`get_genetics_by_study()`</code> , <code>`get_mutations_by_study()`</code> , <code>`get_cna_by_study()`</code> , <code>`get_fusions_by_study()`</code> , <code>`get_clinical_by_study()`</code> , <code>`available_samples()`</code> , <code>`available_patients()`</code> |
| Molecular Profiles | <code>`get_genetics_by_study()`</code> , <code>`get_mutations_by_study()`</code> , <code>`get_cna_by_study()`</code> , <code>`get_fusions_by_study()`</code>                                                                                                                                                                                                                                              |
| Patients & Samples | <code>`get_genetics_by_sample()`</code> , <code>`get_mutations_by_study()`</code> , <code>`get_cna_by_sample()`</code> , <code>`get_fusions_by_sample()`</code> , <code>`get_clinical_by_sample()`</code> , <code>`get_clinical_by_patient()`</code> , <code>`get_panel_by_sample()`</code> , <code>`get_samples_by_patient()`</code>                                                                     |



# MSK IMPACT – How Data is Organized in cBioPortal

You can also pull other sample-level or patient-level characteristics like: age, sex, TMB, cancer type, etc.



| Level              | Functions                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Database           | <code>`available_studies()`</code> , <code>`available_profiles()`</code> , <code>`available_gene_panels()`</code> , <code>`get_genes()`</code>                                                                                                                                                                                                                                                            |
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# MSK IMPACT – data niggling details



IMPACT ID format:

- P-0000113-T01-IM3
- Patient ID-Tumor sample number-IMPACT panel version
- Samples with other prefixes were not sequenced by cBioPortal

Number of samples in study may not correspond to number of samples in genetic data

- Some samples may not have any mutations or copy number alteration

Data formats may vary

- Differences in mut vs cna data - Tumor\_sample\_barcode vs SampleId
- CNA often in wide format and may need transformation

Mutation\_Status

- Categories: GERMLINE, SOMATIC, UNKNOWN. In many cases, UNKNOWN are of interest. Be sure to filter out germline unless you are specifically studying these.

## Impact panels

- IM3 - 341 genes
- IM5 - 410 genes
- IM6 – 468 genes
- IM7 – 505 genes

Example scripts will be available:

<https://github.com/MSKCC-Epi-Bio/introduction-to-msk-impact>



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# MSK IMPACT – gnomeR

- Documentation: <https://mskcc-epi-bio.github.io/gnomeR/>
- gnomeR Slides: [https://genie-bpc.github.io/intro\\_to\\_genieBPC\\_and\\_gnomeR/geniebpc-gnomer-r-med.pdf](https://genie-bpc.github.io/intro_to_genieBPC_and_gnomeR/geniebpc-gnomer-r-med.pdf)





# Introduction to Genomics & MSK-IMPACT data

Thanks for  
your attention!



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